

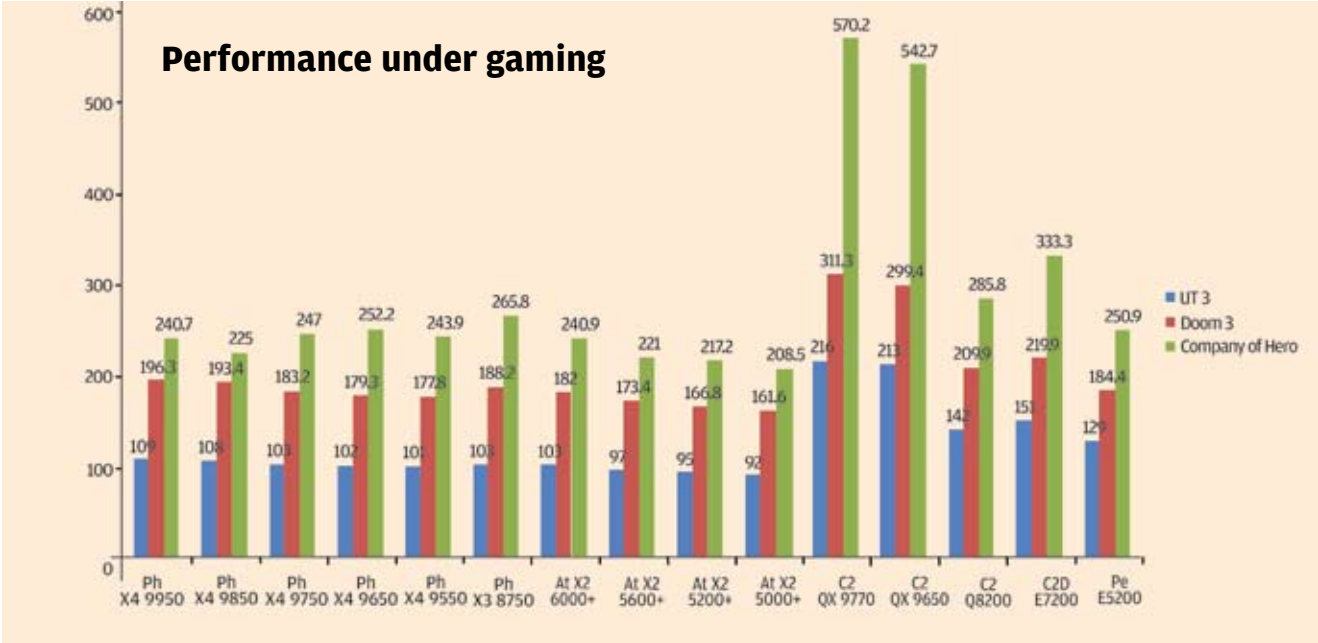
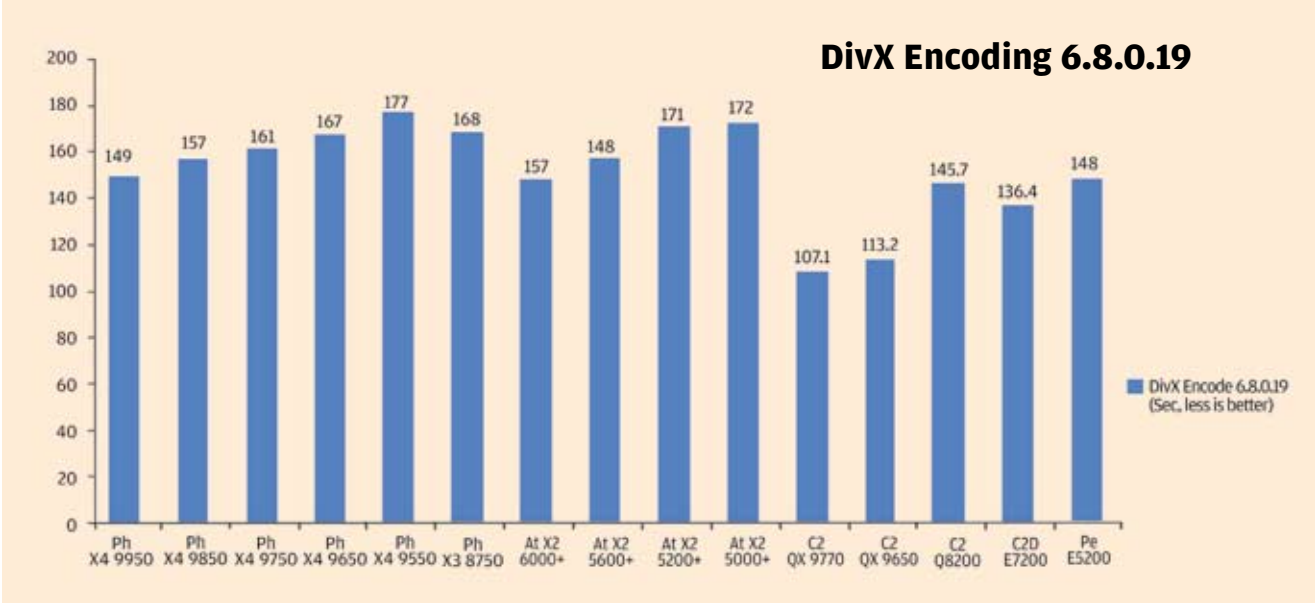
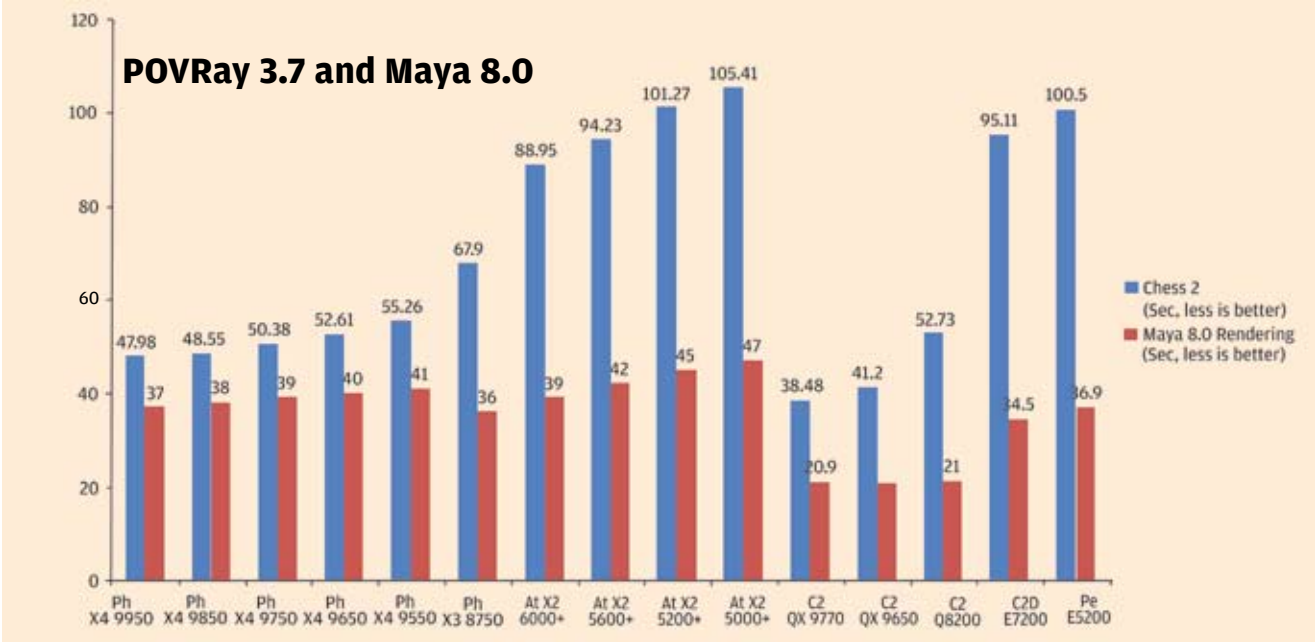
CineBench’s performance numbers. Intel’s three quad cores take the first three positions; the QX9770 and QX9650 are alone at the top and sail past the 10,000 point barrier, which proves too much for all the other CPUs. Even the Q8200 is ahead of the competition by a healthy 800 points or so. Intel’s dual cores are also faster than AMDs alternatives, except the X2 6000+ which edges out the E5200. Anyone looking to use CPUs for any sort of 3D content creation should take a look at Intel’s CPUs as their performance in CPU-intensive rendering features like reflections, ambient occlusion, lighting and

procedural shaders is unmatched. Obviously there is a price angle to be considered as well, but in general, an Intel quad-core will outpace any AMD CPU, as the Q8200 is one of the slowest Intel quad cores around.

More 3D content creation:
POVRay 3.7 Beta and Maya 8.0

Both these rendering benchmarks stress just about any CPU out there. Maya 8.0 has very strong multi-core abilities not to mention it is one of the most popular 3D packages worldwide, along with *3D Studio Max*. POVRay has long been acclaimed for its ability

to stress a CPUs ability to render. Most animators and 3D render artists would want to take a very close look at these scores, along with the score of each CPU in CineBench R10. People with a budget will be looking for best bang for their buck; others will just want the fastest solution around. From the lack of skyscrapers in the Intel corner it’s very easy to tell these CPUs thrive on 3D rendering and content creation. The Intels Q8200 makes a sensible buy for anyone looking to use Maya, as it scores the same as the QX9650 which is much costlier. For some reason AMD doesn’t do well in Maya, even Intel’s low-end



dual cores are faster, this may be in case Autodesk has optimised Maya for use with Intel processors. Under POVRay, AMD’s fastest X4 Phenom’s can hold their own against the slightly costlier Q8200, although if it’s a price no bar situation then Intel wins with its QX9770 and QX9650 duo.

Compression and Encoding:
WinRAR 3.8 And DivX 6.8.0.19

Under WinRAR 3.8 Intel’s QX 9770 and QX 9650 take first and second position followed all five AMD X4 Phenom CPUs. For some reason the 3.0 GHz QX 9650 is even faster than its 3.2 GHz sibling, the QX 9770. The Q8200 is too meek to play hardball with the big boys from AMD – and at 20 per cent less performance but a fifth of the price of the two costliest Intel CPUs, the Phenom X4 9950 makes the most sensible choice for those working with a lot of compression/decompression algorithms. DivX isn’t multi-core optimised at all; we saw CPU utilisation at a steady 25 per cent under Windows’ Task Manager’s Performance pane. This means that only a single core is working and in case of a quad-core CPU the other three cores are sitting idle. This version of DivX is also SSE 4 optimised, which means it is biased towards Intel’s Wolfdale and Yorkfield CPUs; which is why

the lack of multi-threading support came as a shock. We were surprised to see AMDs Phenom X4 9950 and X2 6000+ give Intel’s Q8200 a good run for its money. The two Extreme Edition Intel CPUs are untouchable; but for their prices they had better be.

For gamers:
UT 3, Doom 3 And Company Of Heroes

Today games are the numero uno reason why most of us home users buy fast GPUs and CPUs. Everybody wants their games to look as close to real as possible and words like realism and immersiveness are becoming regular lingo for any gamer. Developers know this, NVIDIA and ATI know this and so do Intel and AMD. Unfortunately games like *Crysis* and *World In Conflict* stress out GPUs more than they stress CPUs and the fastest cards barely choke up 60 fps (30 fps for *World In Conflict*). Such benchmarks are therefore not very suitable for testing CPUs. From amongst our trio of games, *Company Of Heroes* was optimised for Intel CPUs and sometimes this becomes very visible with the tremendous increase in fps over the AMD CPUs. This should be taken with a pinch of salt however, because the difference will not be as noticeable at higher resolutions and higher

visual settings. For purposes of testing CPUs we intentionally keep settings and resolutions low to allow the processors to scale in performance.

Company Of Heroes threw up some weird scores – not only was a slower clocked AMD tri-core (the Phenom X3 8750) faster than the top-end Phenom X4 9950 but Intel’s Q8200 was significantly slower than the cheaper by half E7200. The QX 9770 and QX 9650 remain untouchable across all games. Sadly AMD loses out in *UT 3* – developer support does really matter, and for most game developers jumping on to the leading bandwagon seems the most prudent choice. *Doom 3* was a fantastic engine to benchmark and seeing Intel’s superiority here just cements their claims; developer support be hanged – the Core 2 Duos and quads are the best CPUs around for discerning gamers or anyone who wants maximum eye-candy in their games.

Who Won, Who Lost?
Did we ever use the phrase *you pay for what you get* before? Intel just thrashes AMD at the top-end and the Phenom X4s have no answer to the QX 9770 and the QX 9650 when it comes to brute force processing. With 12 megabyte L2 caches and clock speeds of up to 3.2 GHz these CPUs are the fastest desktop processors available on